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SUPERSEDING
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MILITARY SPECIFICATION

INDICATORS, PROPELLANT TEMPERATURE: M13, M15 AND M33 (LESS PROPELLANT)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers three types of propellant temperature indicators for use in indicating the temperature of the propellant in 75 MM, 90 MM, and 120 MM ammunition (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing)

PPP-B-601 — Boxes, Wood, Cleated Plywood

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner

PPP-B-636 — Box, Fiberboard

PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant

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MIL-P-116 — Preservation, Methods of

MIL-T-3618 — Thermometer, Self-Indicating, Bimetallic: M1A1

MIL-F-13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-I-45208 — Inspection Requirements; General Specification for

STANDARDS

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MIL-STD-105 — Sampling Procedures

FSC 1290

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and Tables for Inspection by Attributes

MIL–STD–129 — Marking for Shipment and Storage

DRAWINGS

MUNITIONS COMMAND

D7689002 — Indicator, Propellant Temperature: M13

D7689006 — Indicator, Propellant Temperature: M15

F8572207 — Indicator, Propellant Temperature: M33

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The Propellant Temperature Indicator: M13 shall be manufactured in accordance with Drawing D7689002; Propellant Temperature Indicator: M15 in accordance with Drawing D7689006; Propellant Temperature Indicator: M33 in accordance with Drawing F8572207, and all drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired characteristics, performance and permanence to insure inherent qualtiy and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL–F–13926:

(a) Order of precedence

(b) Dimensions and tolerances

(c) Inorganic protective surface finishes

(d) Part identification and marking

(e) Workmanship

3.4 Environmental.

3.4.1 *Storage temperature.* The indicator shall show no evidence of physical failure and shall meet all performance requirements of this specification at the standard ambient temperature of plus 15° C. (plus 60° F.) to plus 32° C. (plus 90° F.) after having been exposed and thermally stabilized at ambient temperatures of plus 71.1° C. (plus 160° F.) to minus 62.2° C. (minus 80° F.).

3.4.2 *Vibration.* The indicator shall meet all requirements herein after being vibrated for five minutes plus or minus fifteen seconds at an amplitude of 1/16 inch (⅛ inch total movement) at a frequency of 30 cycles per second.

3.4.3 *Rain.* There shall be no evidence of water penetration or damage to the indicator when subjected to a simulated rainfall of a minimum of 4 inches per hour for a period of 2 hours.

3.5 Window. The window shall be in accordance with its respective drawing listed below, and shall not be adversely affected as a result of being subjected to the requirements of 3.4.1 and 3.4.2:

A7574025	Indicator M13
A7574040	Indicator M15
B8572213	Indicator M33

3.6 Thermometer, Self-Indicating, Bimetallic: M1A1. The Thermometer, Self-Indicating-Bimetallic: M1A1, which forms a part

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of the indicator, shall have met the requirements of MIL–T–3618.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Except where otherwise specified herein, the contractor's inspection system shall be in accordance with MIL–I–45208.

4.3 Lot formation. When inspection lot sizes and lot formations are applicable, they shall be in accordance with MIL–STD–105 and shall also be defined in the contractor's inspection system.

4.3.1 *Defective units and lots.* Defective units and lots shall be processed as specified in MIL–STD–105 and shall not be included as a part of the quantity specified in the contract until formal approval is obtained.

4.4 Classification of defects and sampling plans. Examination and testing to determine acceptance or rejection in compliance with the applicable requirements of Section 3 shall be conducted on a 100 percent inspection basis, initially. After conclusive evidence of the process average has been established in conformance with MIL–STD–105 and the AQL's specified herein, the classification of defects and sampling plans specified in tables I and II may be used in lieu of 100 percent inspection. The AQL value specified shall apply to each individual characteristic on a single defec basis.

4.4.1 *Disposition of rejected lots.* Rejected lots shall be screened in conformance to 4.3.1 for all defective characteristics causing rejection of the lot. After the Government inspector has approved the corrective action taken, subsequent production lots may be submitted for acceptance in accordance with 4.4 and MIL–STD–105.

TABLE I. *Classification of defects*

Use inspection level 7, in table III–B with sampling plan table IV–A of MIL–STD –105.

Critical: None defined

Major	AQL 4.0 percent defective	Requirement paragraph	Q.A. paragraph	Test method
101	Storage temperature	3.4.1	4.6.1.1	Standard measuring equipment (SME).
102	Vibration	3.4.2	4.6.1.2	Special testing equipment (STE).
103	Rain	3.4.3	4.6.1.3	STE
104	Window	3.5	4.6.3	

Minor: None defined

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TABLE II. *Classification of defects*

Use inspection level II, in table III-A with sampling plan table IV-A of MIL-STD-105.

Critical: None defined

Major	AQL 1.5 percent	Requirement paragraph	Test method
101	Unit packaging; improper method (Level A, C)	5.1.1.1 5.1.2.1	MIL-P-116
102	Item improperly immobilized, box not conforming to specification (Level A, C).	5.1.1.1.1 5.1.2.1.1	PPP-B-636
103	Unit packages; incorrect quantity, container not conforming to specification, excess weight (Level A, B, C).	5.2.1 5.2.1.1 5.2.2 5.2.2.1	PPP-B-601 PPP-B-621 Visual SME
104	Markings incorrect, omitted, illegible, not conforming to standard (Level A, B, C).	5.3	MIL-STD-129
Minor	**AQL 4.0 percent**	**Requirement paragraph**	**Test method**
201	Improper tape application, not conforming to specification (Level A).	5.1.1.2 5.1.2.1.2	Visual
202	Workmanship	5.5	Visual

4.5 Inspection equipment.

4.5.1 *Contractor provisions.* Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.5.2 *Equipment to test environmental requirement.* The test equipment and accuracy controls shall be in accordance with the "Test Facilities" requirements specified in MIL-F-13926.

4.6 Test methods and procedures.

4.6.1 *Environmental.*

4.6.1.1 *Storage temperature.* The indicator shall be placed in a suitable test chamber capable of temperature requirements of 3.4.1. The test shall be conducted in accordance with Procedure II of MIL-F-13926. After exposure, the indicator shall meet the requirements of 3.4.1.

4.6.1.2 *Vibration.* The indicator shall be vibrated by means of a vibrating machine capable of vibrating the indicator at the number of cycles and amplitudes indicated in 3.4.2. After vibration, the indicator shall meet all requirements in accordance with this specification.

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4.6.1.3 *Rain.* The indicator shall be placed in a suitable test chamber capable of simulated rainfall in accordance with 3.4.3. The test shall be accomplished in accordance with Procedure I of MIL–F–13926. After completion of test, the indicator shall be visually inspected for the requirements of 3.4.3.

4.6.2 *Thermometer, Self-indicating, Bimetallic: M1A1.* The thermometer indicated in 3.6 shall be inspected in accordance with the test methods and procedures specified in MIL–T–3618.

4.6.3 *Window.* After having been subjected to the requirements of 3.4.1 and 3.4.2, the window shall be visually inspected, and shall be transparent, clear, and show no evidence of damage in accordance with the requirements of 3.5.

4.7 Preservation and packaging.

4.7.1 *Preproduction pack.* The preproduction pack (specified in 5.4) shall be inspected 100 percent by the contractor or his representative, and witnessed by the Government inspector to determine conformance with the requirements of this specification for Level A.

4.7.2 *Preproduction pack failure.* Should the preproduction pack fail to pass the test for Government acceptance, packaging operations will cease until defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new pack shall be completely inspected by the contractor and witnessed by the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* The item, in its fiber tube, shall be packaged in accordance with Method IC–5 of MIL–P–116.

5.1.1.1.1 *Immobilizing in fiberboard box.* The item shall be immobilized in a fiberboard box conforming to Specification PPP–B–636, type I, class 2.

5.1.1.1.2 *Application of tape.* The fiberboard box shall be sealed by the application of tape conforming to Specification PPP–T–76.

5.1.2 *Level C.*

5.1.2.1 *Unit packaging.* The item, in its fiber tube, shall be packaged in accordance with Method III of MIL–P–116.

5.1.2.1.1 *Immobilizing in fiberboard box.* The item shall be immobilized in a fiberboard box conforming to Specification PPP–B–636, type I, class 1.

5.1.2.1.2 *Application of tape.* The box shall be closed by the application of tape conforming to Specification UU–T–111.

5.2 Packing.

5.2.1 *Level A.* A quantity of unit packages shall be packed in a shipping container conforming to either of the following specifications:

PPP–B–601 — Oversea type

PPP–B–621 — Class 2

The gross weight of the packed shipping container shall not exceed 200 pounds.

5.2.1.1 *Exception to wood shipping container.* When the quantity requisitioned or shipment consists of one unit, the fiberboard box constituting the unit container shall be used as the shipping container.

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5.2.2 *Level* B. A quantity of unit packages shall be packed in a shipping container conforming to either of the following specifications:

PPP–B–601 — Domestic type.

PPP–B–621 — Class 1

The gross weight of the packed shipping container shall not exceed 200 pounds.

5.2.2.1 *Exception to wood shipping container.* When the quantity requisitioned or shipment consists of one unit, the fiberboard box constituting the unit container shall be used as the shipping container.

5.2.3 *Level C.* Packing shall be as specified for Level B.

5.3 **Marking.** Interior packages and exterior containers shall be marked in accordance with MIL–STD–129.

5.4 **Preproduction pack.** Prior to quantity packaging, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the requirements of this specification.

5.5 **Workmanship.** Tape shall be uniformly applied to interior boxes and shall completely cover and adhere to all joints and seams. Shipping containers shall be free of protruding staples, screws, nails, etc. Strapping shall be applied just prior to shipment and shall not cover any container marking. Irrelevant marking shall be neatly obliterated. Visual observation shall indicate that the application of specified materials and methods are practical and will not result in packaging or packing deficiencies.

6. NOTES

6.1 **Intended use.** Propellant temperature indicators covered by this specification are used for indicating propellant temperature for the following type ammunition:

M13 — 90 MM ammunition

M15 — 120 MM ammunition

M33 — 75 MM ammunition

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable stock number.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder, or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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